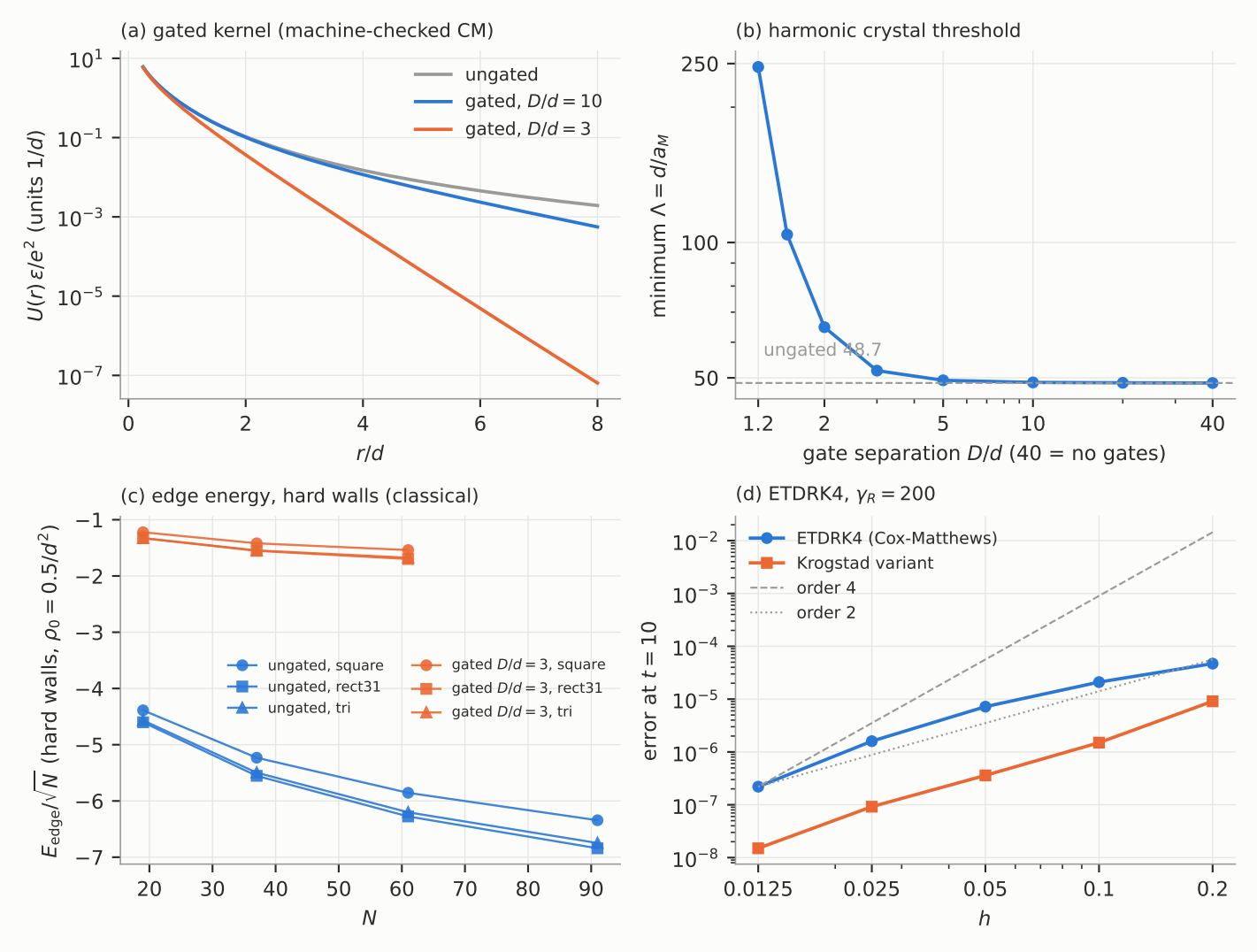
(a) gated kernel (machine-checked CM)
ungated
gated, D/d = 10
gated, D/d = 3
U(r) ε/e² (units 1/d)
r/d
(b) harmonic crystal threshold
minimum Λ = d/a_M
ungated 48.7
gate separation D/d (40 = no gates)
(c) edge energy, hard walls (classical)
ungated, square
ungated, rect31
ungated, tri
gated D/d = 3, square
gated D/d = 3, rect31
gated D/d = 3, tri
E_edge/√N (hard walls, ρ0 = 0.5/d²)
N
(d) ETDRK4, γR = 200
ETDRK4 (Cox-Matthews)
Krogstad variant
order 4
order 2
error at t = 10
h